A REFORMULATION OF THE G-RING CONDITION THROUGH THE FORMAL FIBRES OF QUOTIENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- G-ring condition and geometrically regular formal fibres. definition environment at source lines 12179-12192 and the discussion immediately before the counted statement, source lines 12185-12192. Supplies both sides of the equivalence: the condition on local rings and the condition on the algebras displayed in the statement.
- Completion of a local ring commutes with passing to a quotient by a prime contained in it. used at proof lines 12204-12209, in the displayed identification of the two algebras. Is the computation that makes the two families of algebras over the residue field the same.
- Base change of the residue field extension along the quotient map. used at the same display, where the factor over kappa(q) is written on both sides. Shows that the two algebras are compared over the same field, so geometric regularity of one is equivalent to geometric regularity of the other.

Source reference. Original lemma 07PN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9803–9803. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a Noetherian ring. Then R is a G-ring if and only if for every pair of primes $\mathfrak{q} \subset \mathfrak{p} \subset R$ the algebra*

$$(R/\mathfrak{q})^\wedge_\mathfrak{p} \otimes_{R/\mathfrak{q}} \kappa(\mathfrak{q})$$

is geometrically regular over $\kappa(\mathfrak{q})$.

Proof. This follows from the fact that

$$R^\wedge_\mathfrak{p} \otimes_R \kappa(\mathfrak{q}) = (R/\mathfrak{q})^\wedge_\mathfrak{p} \otimes_{R/\mathfrak{q}} \kappa(\mathfrak{q})$$

as algebras over $\kappa(\mathfrak{q})$. □